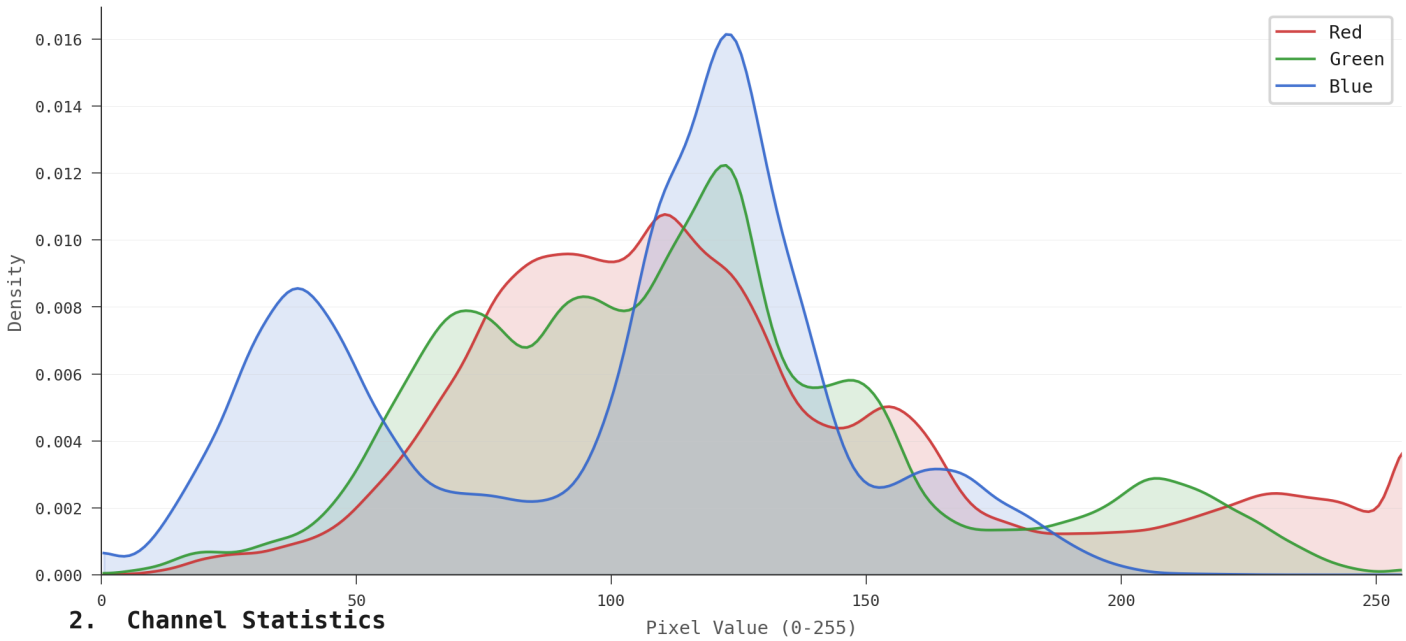
Reference Data Sheet: KODIM19

Subject: Lighthouse and Coast

512x768 | 393,216 px
24-bit RGB | sRGB | PNG (lossless)

1. RGB Channel Distribution



2. Channel Statistics

Channel	Mean	Std Dev	Variance	Kurtosis	Skewness	Min	Max
Red	122.4976	51.1987	2621.3117	0.2442	0.8550	0	255
Green	115.5019	46.2530	2139.3377	0.0530	0.5868	0	255
Blue	98.1700	45.4169	2062.6919	-0.9406	-0.2592	0	255

3. Inter-Channel Correlation

	R	G	B
R	1.0000	0.9668	0.8155
G	0.9668	1.0000	0.9151
B	0.8155	0.9151	1.0000

Pairwise Pearson Correlation Coefficients

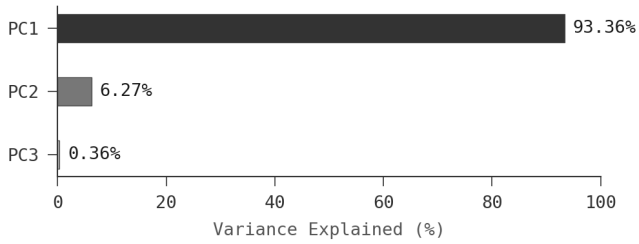
r(R-G)	=	**0.966830**
r(R-B)	=	**0.815462**
r(G-B)	=	**0.915071**
Avg |r|	=	**0.899121**

4. Covariance Matrix S

$$
\begin{array}{c} \\ R \\ G \\ B \end{array}
\begin{array}{c}
\begin{array}{ccc} R & G & B \end{array} \\
\begin{bmatrix}
\mathbf{2621.3184} & 2289.5494 & 1896.1883 \\
2289.5494 & \mathbf{2139.3431} & 1922.2630 \\
1896.1883 & 1922.2630 & \mathbf{2062.6971}
\end{bmatrix}
\end{array}
$$

Reference Data Sheet: KODIM19

5. Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	**-0.6208**	-0.5770	-0.5307	6370.5766
PC2	-0.6120	-0.0664	0.7881	428.1205
PC3	0.4900	-0.8140	0.3119	24.6615

6. Derived Metrics

Condition Number

258.32

(l1/l3) PC1/PC2: 14.88 | PC2/PC3: 17.36

Blue Channel Independence

13.0%

Fraction of blue variance not captured by PC1

PC1 Dominant Channel

Red

Loading: -0.6208 | Red dominant

Classification: Strongly One-Dimensional (PC1 = 93.36%)

7. Spatial Autocorrelation (lag-1)

Channel	Horizontal	Vertical	Average
Red	0.934066	0.952822	0.943444
Green	0.921641	0.941917	0.931779
Blue	0.926385	0.945151	0.935768

8. Average Local Variance (3x3)

Channel	Value
Red	247.2722
Green	239.9454
Blue	211.1540

9. Redundancy Profile

Classification: Strongly One-Dimensional | Condition Number: 258.32 | PC1: 93.36%

Correlation: R–G pair dominates at r = 0.966830 | G–B: 0.915071 | R–B: 0.815462

Blue Channel: 13.0% independent | Moderate decoupling from PC1

Spatial: Avg autocorrelation: 0.936997 | Avg local variance: 232.7905

Eigenvector: Red dominant | R (-0.6208) leads | G (-0.5770), B (-0.5307)

References

[1] Eastman Kodak Company. Kodak Publication No. PCD-042, 1992.

[2] Baetzel, J. (2026). "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."

[3] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.

[4] Giorgianni, E.J. and Madden, T.E. Digital Color Management. Addison-Wesley, 1998.

Blue Channel Independence = (1 - (blue_loading_PC1^2 x l1 / Var(B))) x 100

Data Source: Statistics derived from Baetzel (2026), "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."